

Daive Macario

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Skills

Technical	Python, Golang, Terraform, AWS, Linux, Bash, Kubernetes, Docker, CI-CD tooling (Git, GitHub Workflows, GitLab CI), Zero-trust, Ansible
Languages	Italian (native), English (C1 - IELTS 8.0), French (basic), Dutch (basic)
Interests	ML/AI applications, Open-Source Software, MLOps, self-hosting/homelab, Distributed Computing, Software Architecture, Edge Computing, IoT, Linux server management

Experience

- Vitestro B.V., DevOps Engineer** | Utrecht, NL Mar 2024 – Current
- Lead effort to achieve integration of product with hospital's Laboratory Information Systems (LIS); deployed and maintained integration engine hosted on AWS (ECS Fargate), including centralized monitoring platform.
 - Implemented solution for uploading relevant data (telemetry, sessions) from the device to AWS, including the use of Fluent Bit and custom tooling for specific data types, enabling near-real-time retrieval of data.
 - Supported the other R&D teams with internal tooling, including data migrations, package registries, container registries, container image development.
- Vitestro B.V., Entry-Level DevOps Engineer** | Utrecht, NL Sept 2024 – Feb 2024
- Member of Platform & Connectivity team working on integrating the world's first **Autonomous Robotic Phlebotomy Device** with local and public cloud.
 - Engineered a full GitLab CI ecosystem on AWS, including autoscaled runners, Docker registry mirror, repository caches; achieved costs reduction compared to existing setup.
 - Contributed to the development and maintenance of custom Embedded Linux distribution (**Yocto, Bitbake**).
- University of Illinois at Chicago, Graduate Research Assistant** | Illinois, USA Jan 2024 – Jun 2024
- Research work focusing on distributed computing applied to **Machine Learning** and **Generative AI** models.
 - Designed and implemented a framework for deploying state-of-the-art LLMs on networks of constrained edge devices.
 - Won **Best Paper Award** at IEEE LANMAN 2025 for the work on "A Model-Distributed Inference Approach for Large Language Models at the Edge".

Education

- University of Illinois at Chicago, MS in Electrical and Computer Engineering** | IL, USA GPA: **4.0 / 4.0** July 2024
Double-degree program in collaboration between Politecnico di Torino and University of Illinois at Chicago.
- Politecnico di Torino, MS in ICT for Smart Societies** | Torino, Italy GPA: **30.0 / 30** July 2024
- Politecnico di Torino, BS in Electronics and Communications Engineering** | Torino, Italy GPA: **29.7 / 30** Sept 2022
- M.Sc. Thesis:** "A Model-Distributed Inference Approach for Large Language Models at the Edge".
- Honors:** 5-times winner of scholarships for top students. Took part in "Percorso Giovani Talenti" during B.Sc. Took part in "Alta Scuola Politecnica (ASP)" during M.Sc. Winner of scholarship for double-degree program (TOP-UIC).
- Associations:** IEEE Eta-Kappa-Nu (honor society) Mu-Nu chapter member (Politecnico di Torino).

Projects

- DMHosted – Kubernetes-powered Homelab** October 2025 – Onwards
- Bare-metal **Kubernetes** (K3s) cluster running in homelab, running both internal (VPN) and public-facing services
 - GitOps** setup via ArgoCD, centralized **monitoring and observability** (Prometheus) and **Ansible** for provisioning and server upgrades
 - <https://github.com/davmacario/dmhosted-infra>
- A Model-Distributed Inference for Large Language Models at the Edge** | M.Sc. Thesis Jan 2024 – June 2024
- Distributed computing approach to LLM inference over a network of Edge devices (Nvidia Jetson TX2).
 - Implementation of a parallel processing algorithm to achieve increased generation rates by minimizing idle nodes.
 - Ability to run the **GPT-2**, **Llama 3**, and **Mistral** model families achieving usable throughput.
 - <https://github.com/davmacario/MDI-LLM>
- FREISA | OpenCV AI Competition 2023 (Grand prize winner)** July 2023 – December 2023
- "Four-legged Robot Ensuring Intelligent Sprinkler Automation".
 - Computer vision-enabled quadruped robot able to detect plants, approach them and decide whether to sprinkle water on them based on the leaves' condition.
 - Focused on developing the computer vision module (**Docker** container) using an OAK-D Lite camera, **YOLOv8**, **OpenCV**, and **DepthAI**, providing fast APIs to the motion controller (ROS 2).
 - <https://github.com/B-AROL-0/FREISA>

Certifications

Software Architecture for Developers – by Simon Brown (Trifork)
HashiCorp: Terraform Associate (003)

May 2026
Aug 2025